dissonance is the mental discomfort that one feels when new information conflicts with previously held beliefs or cognitions. To resolve this discomfort, people tend to notice only information of interest to them (called selective exposure), ignore or modify information that conflicts with existing beliefs (called selective perception), and/or remember and consider only information that confirms existing beliefs (called selective retention). Aspects of these behaviors are contained in the biases categorized as belief perseverance. The six belief perseverance biases covered in this book are cognitive dissonance, conservatism, confirmation, representativeness, illusion of control, and hindsight.

Information Processing Biases

The second category of cognitive biases has to do with "processing errors," and describes how information may be processed and used illogically or irrationally in financial decision making. As opposed to belief perseverance biases, these are less related to errors of memory or in assigning and updating probabilities and instead have more to do with how information is processed. The seven processing errors discussed are anchoring and adjustment, mental accounting, framing, availability, self-attribution bias, outcome bias, and recency bias.

Individuals are less likely to make cognitive errors if they remain vigilant to the possibility that they may occur. A systematic process to describe problems and objectives; to gather, record, and synthesize information; to document decisions and the reasoning behind them; and to compare the actual outcomes with expected results will help reduce cognitive errors.

EMOTIONAL BIASES

Although *emotion* has no single universally accepted definition, it is generally agreed upon that an emotion is a mental state that arises spontaneously rather than through conscious effort. Emotions are related to feelings, perceptions, or beliefs about elements, objects, or relations between them; these can be a function of reality or the imagination. Emotions may result in physical manifestations, often involuntary. Emotions can cause investors to make suboptimal decisions. Emotions may be undesired to the individuals feeling them, and while they may wish to control the emotion and their response to it, they often cannot.

Emotional biases are harder to correct for than cognitive errors because they originate from impulse or intuition rather than conscious calculations. In other words, a bias that is an inclination of temperament or outlook,